An Ultrawideband 1-Bit Suspended Reconfigurable Intelligent Surface for Enhancing Wireless Coverage

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***Abstract*—This letter proposes an ultrawideband 1-bit suspended reconfigurable intelligent surface (RIS) designed to enhance the signal coverage of wireless communication networks. The RIS unit cell comprises two metal gratings connected via a PIN diode, a dielectric-air-dielectric interlayer, and a metal reflective layer. The RIS achieves a stable reflection phase difference of 180° ± 30° between the ON and OFF states of the PIN diode within the ultrawideband range of 2.75 GHz to 6 GHz. This range includes the 3.4 GHz and 4.9 GHz bands for 5G communication, the 5.2 GHz band for indoor wireless communication, and the 5.8 GHz band for Wi-Fi and Bluetooth signals. A prototype, consisting of 20 × 20 unit cells, was fabricated and measured, demonstrating two-dimensional wide-angle beam scanning of ± 60° within the working frequency range. Furthermore, a RIS-assisted wireless communication system was constructed, confirming that deploying RIS can significantly enhance communication quality in targeted areas by transmitting and receiving image information at different frequencies. The proposed RIS holds substantial potential for RIS-assisted wireless communication systems.**

***Index Terms*—Beam deflection, reconfigurable intelligent surface, ultrawideband, wireless communication.**

I. Introduction

METASURFACES, as two-dimensional equivalents of electromagnetic metamaterials, have garnered significant attention in recent years due to their ability to control the phase, amplitude, and polarization of electromagnetic waves [1]. Generally, the function of passive metasurfaces is fixed after fabrication, limiting their ability to flexibly manipulate electromagnetic waves [2]. To achieve dynamic real-time manipulation, reconfigurable intelligent surfaces (RISs) incorporating tunable components [3] and tunable materials [4] have been extensively studied. This advancement has led to many attractive applications, including stealth technology [5], imaging [6], wireless communication systems [7], and radar systems [8].

With the growing demand for wireless communication, RISs have emerged as an appealing solution for enhancing indoor wireless communication [9]. By dynamically guiding the reflected beam, RISs can cover signal blind spots and improve signal quality in designated areas. In Li et al. [10], proposed a RIS-assisted massive backscatter wireless communication scheme for 2.4 GHz Wi-Fi signals, which reallocates stray wireless wave energy toward desired locations, verified through image transmission experiments. In Cui et al. [11], developed a dual-channel wireless communication system based on space-time-coding RIS, transmitting different images to two users. However, most existing RISs operate on narrow bands [12] or single frequencies [13], [14], limiting their ability to meet wideband wireless communication requirements.

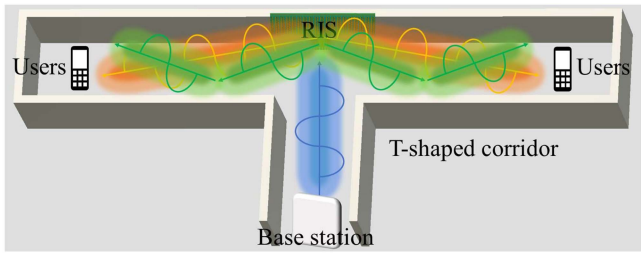


Fig. 1. Schematic of 1-bit RIS application in a T-shaped corridor.

In wireless communication systems, a wider bandwidth is desirable to cover more communication frequency bands. Recently, some wideband RISs have been presented to realize wideband beam manipulation [15], [16], [17], [18], [19], [20]. An ultrawideband low-loss RIS proposed in [15] achieves a 1-bit phase shift from 6.75 GHz to 11.25 GHz and a 50% relative bandwidth. A wideband RIS was designed to provide stable wireless communication service in the frequency range of 3.7 GHz to 5.1 GHz with a relative bandwidth of 31.8% [9]. However, these wideband RISs can only enhance signal coverage in a single or two frequency band, which cannot cover the different frequency bands adopted by the various standards of current wireless communication. Moreover, using a suspended structure in the design of RISs can reduce the magnetic coupling between the surface and ground plane, suppressing the Q-factor of magnetic resonance and further enhancing the bandwidth characteristics of RIS [21], [22].

In this letter, to meet the requirement of ultrawideband dynamic two-dimensional beam manipulation for advanced wireless communications, an ultrawideband 1-bit suspended RIS was designed, fabricated, and measured. The schematic diagram of the ultrawideband 1-bit RIS application in a T-shaped corridor is shown in Fig. 1. The measured results show that the proposed RIS has a stable reflection phase difference of 180° ± 30° from 3 GHz to 6.25 GHz, with a relative bandwidth of 70.3%. Additionally, an ultrawideband 1-bit RIS-assisted intelligent

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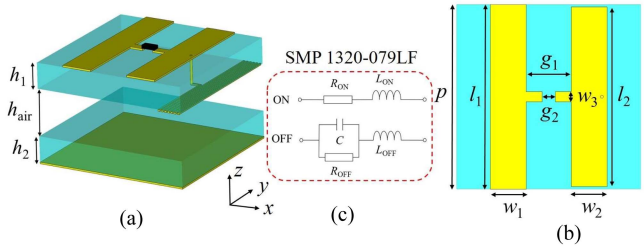


Fig. 2. Schematic of the designed unit cell. (a) Perspective 3-D view. (b) Top metal gratings. (c) Equivalent circuit of PIN diode.

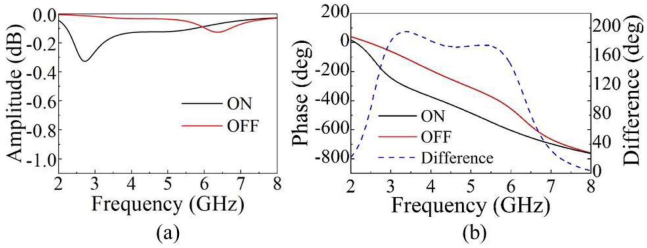


Fig. 3. (a) Reflection amplitude and (b) phase response of the unit cell.

wireless communication system was constructed, verifying the RIS's ability to significantly improve communication quality in the required area through real-time information transmission of different images at various frequencies.

II. Design of the Ultrawideband Suspended RIS

The schematic of the designed suspended RIS unit cell is shown in Fig. 2. It consists of two metal gratings connected through a PIN diode, a dielectric-air-dielectric interlayer, and a metal reflective layer. The pattern layer is suspended above the bottom reflective layer through an air gap to achieve wideband characteristics. The right metal gratings are connected to the bias lines on the back of the first dielectric layer through vias to allow independent control of the PIN diodes. The influence of bias lines on the electromagnetic response of the unit cell has been considered in the simulation. The length of the left gratings equals the unit cell period, so the left gratings of the whole column are connected as the common ground of the PIN diodes. The unit cell period is $p = 18$ mm, approximately equal to 0.26λ at the center frequency. The thickness of the dielectric layers and the air layer are $h_{1(2)} = 3$ mm and $h_{\text{air}} = 5.5$ mm, respectively. The relative permittivity and loss tangent of the dielectric substrate are 2.5 and 0.001, respectively. The other detailed parameters are as follows: $w_{1(2)} = 3.5$ mm, $l_1 = 18$ mm, $l_2 = 17.5$ mm, $g_1 = 4.4$ mm, $g_2 = 1.3$ mm, and $w_3 = 1$ mm. The PIN diode used in the unit cell is the Skyworks SMP1320-079LF, and its equivalent circuit is given in Fig. 2(c), where $R_{\text{ON}} = 0.5\ \Omega$, $L_{\text{ON}} = 0.7$ nH, $C = 0.24$ pF, $R_{\text{OFF}} = 0.4$ MΩ, and $L_{\text{OFF}} = 0.5$ nH.

The unit cell was simulated using periodic boundaries and a Floquet port in CST Microwave Studio. The reflection amplitude and phase response of the unit cell between the ON and OFF states of the PIN diode are shown in Fig. 3. The simulated results indicate that the unit cell has distinct resonant frequencies in the ON and OFF states of the PIN diode, with a maximum loss of 0.33 dB within the frequency band. The widely separated resonant

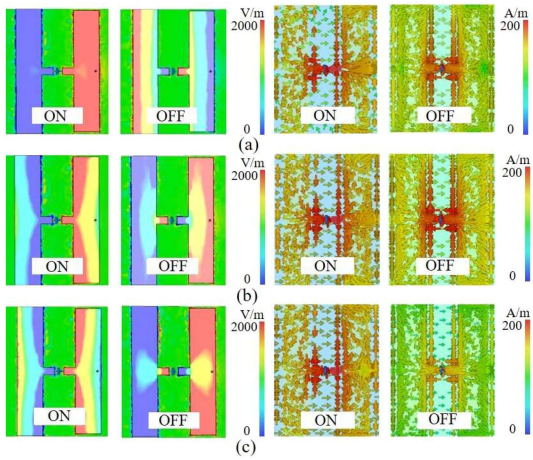


Fig. 4. Electric field distribution and the surface current distribution of the unit cell at the frequency of (a) 3.4 GHz, (b) 4.9 GHz, and (c) 5.8 GHz.

frequencies of 2.7 GHz and 6.4 GHz enable the unit cell to achieve a wideband phase response. Notably, the unit cell can achieve a phase difference of $180° \pm 30°$ in the ultrawideband range of 2.75 GHz to 6 GHz, with a relative bandwidth of 74.3%. These results demonstrate that the designed RIS unit cell exhibits good performance with ultrawideband capabilities and low loss.

To elucidate the mechanism of the ultrawideband phase response of the RIS, the electric field distribution and the surface current distribution of the unit cell in the two states of the PIN diode at frequencies of 3.4 GHz, 4.9 GHz, and 5.8 GHz were simulated, as shown in Fig. 4. The two metal gratings are short-circuited in the ON state, while a capacitance exists between the two metal gratings in the OFF state. In the ON state, the two gratings form an electric dipole to generate electric resonance, whereas in the OFF state, it is equivalent to a double electric dipole resonance. These different resonant modes lead to the generation of an abrupt phase change. Moreover, the surface currents are inverted in both states, resulting in a phase difference of 180°.

According to the pattern multiplication principle, to deflect the beam in the direction of (θ, φ), the phase distribution for the unit cells of the RIS can be expressed as

$$\varphi_{m,n} = -k\left((m-1/2)\,d_x\cos\varphi\sin\theta + (n-1/2)\,d_y\sin\varphi\sin\theta\right) \tag{1}$$

where $k = 2\pi/\lambda$ is the propagation constant, and d_x and d_y are the periodicity of unit cells in x- and y-directions, respectively.

The 1-bit phase quantization of the unit cells can be expressed as

$$\psi_{m,n} = \begin{cases} \frac{\pi}{2}, \varphi_{m,n} + \varphi_0 \in [0, \pi) \\ \frac{3\pi}{2}, \varphi_{m,n} + \varphi_0 \in [\pi, 2\pi) \end{cases} \tag{2}$$

where $\psi_{m,n}$ is the quantized phase, $\varphi_{m,n}$ is the phase calculated theoretically, and φ_0 is the artificially selected additional reference phase for all unit cells.

To verify the dynamic two-dimensional scanning capability of the proposed RIS for incident electromagnetic waves, the radiation patterns of the array composed of 20×20 unit cells with different beam deflection angles at 3.4 GHz, 4.9 GHz, and 5.8 GHz were simulated, as shown in Fig. 5. The radiation

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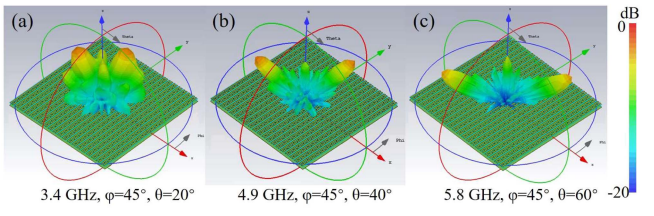


Fig. 5. Radiation patterns of the RIS array with different beam deflection angles at (a) 3.4 GHz, (b) 4.9 GHz, and (c) 5.8 GHz.

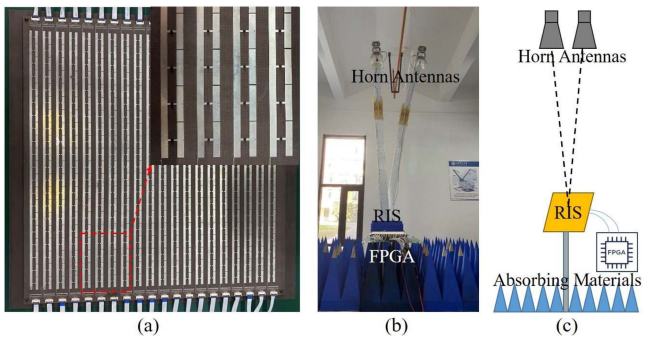


Fig. 6. (a) Fabricated RIS prototype. (b) Image of the experimental system. (c) Schematic of the experimental system.

patterns were simulated by CST using an open boundary and plane wave excitation with different phase distributions. As the deflection angle increases, the performance of beam deflection tends to deteriorate. It can be observed that the designed RIS can deflect incident electromagnetic waves to specific directions between 0° and 60°, demonstrating good beam deflection and scanning performance.

III. Fabrication and Measurement of the Ultrawideband Suspended RIS

To verify the performance of the designed RIS, a prototype was fabricated using printed circuit board technology, as shown in Fig. 6(a). The rectangular array consists of 20 × 20 unit cells measuring 400 mm × 400 mm. The dielectric layers are suspended and fixed by plastic studs at the four corners, forming a dielectric-air-dielectric interlayer. Additionally, 13 nH choke inductors (Murata LQP03HQ13NH02D) are loaded on the dc bias lines to isolate the RF signal. Since the state of the PIN diode of each unit cell needs to be controlled independently, a field-programmable gate array (FPGA) provides different bias voltages. It is worth noting that the total power consumption of the RIS system in operation is 4.8 W to 8.2 W.

The fabricated prototype was measured using the reflectivity experimental system, as depicted in Fig. 6(b) and (c). The experimental system consists of a vector network analyzer, two arched arms loaded with a transmitting horn antenna and a receiving horn antenna, and arched arm rotation control equipment. The left horn antenna radiates a quasi-plane wave as excitation, while the right horn antenna receives the reflected wave from the proposed RIS. The right arm rotates to scan and record the reflection coefficients for different angles to build the radiation patterns. Additionally, absorbent materials were laid on the ground and pointed by the horn antennas to absorb scattered waves.

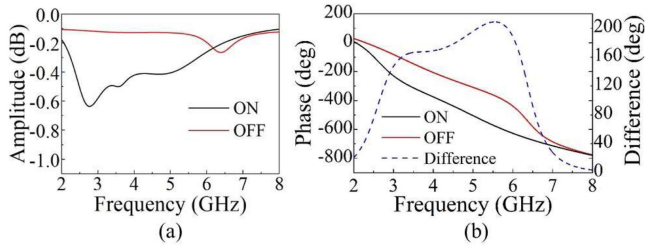


Fig. 7. Measured (a) amplitude response and (b) phase response of the unit cell.

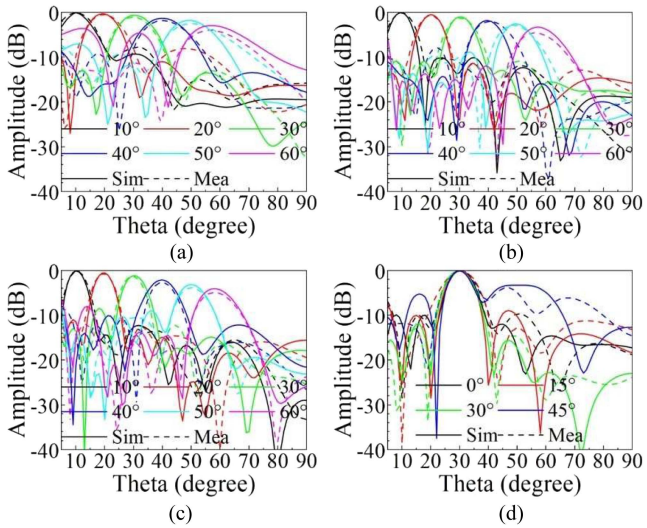


Fig. 8. Simulated and measured radiation patterns of the RIS array at (a) 3.4 GHz, (b) 4.9 GHz, and (c) 5.8 GHz. (d) Simulated and measured radiation patterns with different incident angles at 4.9 GHz.

By setting all PIN diodes to either the ON or OFF state, the amplitude response and phase response were measured in both states, as shown in Fig. 7. Due to limitations of the experimental system, the incident angle of the transmitting antenna is set to 4° to approximate normal incidence, while the receiving antenna is placed symmetrically on the opposite side of the normal plane. The measured results show that the fabricated RIS meets the ultrawideband 1-bit phase response within 3 GHz to 6.25 GHz, with a relative bandwidth of 70.3% and a maximum loss within the frequency band of 0.57 dB. It can be observed that the amplitude response slightly decreases compared to the simulation, and the phase response has a slight deviation. The primary reasons for the deviation include manufacturing errors of the prototype, measurement errors of the system, and the bending of the RIS plane under gravity due to insufficient rigidity and lack of plastic studs supporting the top layer at the center.

In subsequent experiments, we measured the dynamic real-time beam deflection performance of the RIS prototype. Due to the limitations of the experimental system, we provided the one-dimensional beam scanning results of the RIS in the *xoz*-plane. By setting different phase codes into the RIS, the radiation patterns of the beams with different deflection angles at 3.4 GHz, 4.9 GHz, and 5.8 GHz were obtained, as shown in Fig. 8. The main beams all point to specific directions, verifying the correctness of the theoretical design. As the beam deflection angle increases, the beam deflection performance deteriorates. Additionally, considering the application scenario of oblique

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Fig. 9. Schematic of RIS-assisted intelligent wireless communication system.

incidence, we provided the radiation patterns of the incident wave at different incidence angles with the beams pointed in specific directions, as shown in Fig. 8(d). It can be seen that the proposed RIS can still maintain good beam direction under 45° oblique incidences.

IV. RIS-Assisted Wireless Communication System

In this section, we constructed a RIS-assisted intelligent wireless communication system and verified that the RIS can enhance the quality of wireless communication. The system consists of a transmitting horn antenna, a receiving horn antenna, a software-defined radio device (ANTSDR-E310), a RIS, and host computers, as shown in Fig. 9. The transmitting horn antenna is placed 1.2 m from the RIS, with the incident wave direction aligned along the normal direction of the RIS. The receiving horn antenna is positioned at a 45° angle to the normal of the RIS. By controlling the FPGA module through another host computer, the phase coding data is written in real time to deflect the incident wave in the desired direction. Thus, the blind area coverage of the wireless communication network is enhanced, and the communication quality in the required area is improved. The original image information can be obtained by demodulating the received signal.

For verification, we selected the 3.4 GHz and 4.9 GHz bands of 5G communication, as well as the 5.8 GHz band of Wi-Fi and Bluetooth signals, to transmit and receive different image information. Table I illustrates the original images sent at different frequencies using phase shift keying (de)modulation, the images received with the RIS switched OFF and reflecting the incident wave in the normal direction, and the images received after the RIS deflects the incident wave toward the receiving horn antenna. It can be observed that image transmission errors occur when the RIS is not functioning, resulting in image distortion. Conversely, the communication quality at the specific location is significantly improved when the RIS is operational, verifying the RIS's effectiveness in enhancing indoor wireless communication. The relatively good images received when the RIS is switched OFF are due to the side lobes of the transmitting antenna and scattered waves contributing to the propagation link, allowing part of the wireless signal to be received by the receiving antenna.

Table II summarizes the comparison of the design performance of the proposed RIS with some previous related works.

TABLE I
Comparison of Images in Different Situations

Carrier Frequency	Original Image	Received Image Without RIS	Received Image With RIS
3.4 GHz			
4.9 GHz			
5.8 GHz			

TABLE II
Performance Comparison With Some Reported Works

Ref.	Working Frequency (GHz)	Relative Bandwidth	Beam Deflection Range	Wireless Communication Verification
[9]	3.7–5.1	31.8%	± 50°	Yes
[12]	3.5–3.8	8.2%	± 50°	Yes
[15]	6.75–11.25	50%	Not Given	No
[17]	7.71–9.43	20%	Not Given	No
[18]	8–13	47.6%	± 50°	No
This Work	2.75–6	74.3%	± 60°	Yes

Our design achieves a stable phase difference of 180° ± 30° within a maximum relative bandwidth of 74.3% and has the maximum beam deflection range. The proposed suspended RIS exhibits excellent ultrawideband and beam deflection properties.

V. Conclusion

In this letter, a 1-bit suspended RIS with an ultrawideband phase response was designed, fabricated, and measured. The measured results show that the RIS can achieve dynamic beam deflection with a scanning range of ± 60° within the working frequency. The measured results are in good agreement with the simulation results, verifying the performance of the proposed RIS. Moreover, a RIS-assisted wireless communication system was constructed for experimental verification. The feasibility and effectiveness of RIS for enhancing wireless communication quality have been demonstrated. The proposed RIS offers promising application prospects in the new generation of intelligent wireless communication systems.

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